

REVEAL: REal and Virtual Environments Augmentation Lab @ Bath

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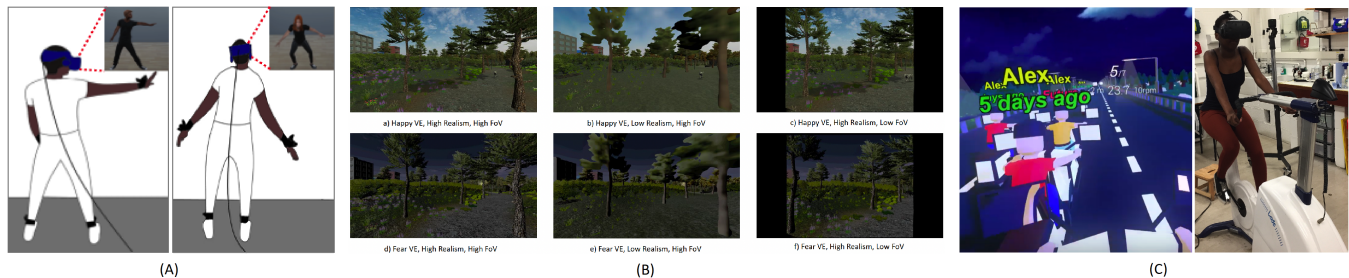


Figure 1: This demo showcases three CHI papers from University of Bath's REVEAL research centre spanning (A) Avatars, (B) Presence, and (C) Exergaming in Virtual Reality

ABSTRACT

The REal and Virtual Environments Augmentation Lab (REVEAL) is a research centre for Human-Computer Interaction at the University of Bath with a focus on immersive technology. Virtual reality (VR) has the potential to change the way we experience the world and interact with each other. It can be used for a wide range of applications such as exercise, training, entertainment, and therapy. In this demo paper, we will showcase some of our VR research that has been published at CHI which can enhance VR experiences across three specific areas of study: learning with avatars, presence, and exergaming.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI).**

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KEYWORDS

Virtual Reality, Avatars, Learning, Presence, Exergaming

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1 INTRODUCTION

The REal and Virtual Environments Augmentation Lab (REVEAL) is a research centre for Human-Computer Interaction at the University of Bath. REVEAL is an interdisciplinary group of researchers from Computer Science and Psychology working at the interface of virtual, augmented and physical experiences. Virtual reality (VR) has the potential to change the way we experience the world and interact with each other, and allows users to experience out-of-this-world experiences from the safety of their home. In this interactivity demo, we will demonstrate how our research can enhance VR experiences by focusing on three research themes: learning with avatars, presence, and exergaming.

2 AVATARS IN VR - DANCING WITH THE AVATARS: MINIMAL AVATAR CUSTOMISATION ENHANCES LEARNING IN A PSYCHOMOTOR TASK [16]

Acquisition of psychomotor skills is commonly achieved through observational learning: a task is learned by observing someone else perform it [24]. At the heart of observational learning in virtual environments are *avatars*: animated digital representations of an instructor that demonstrate a task [48]. With the increasing popularity of virtual environments for learning – both through screens and VR headsets – and the advent of ‘metaverse’ technology, avatars are becoming more and more commonplace in training and education [33]. Avatars can take many forms, with varying degrees of realism, which has been shown to affect user behaviour and cognition (e.g., [49]). This crucial role of avatars raises the question of how instructor avatars should be designed, and whether their design can enhance psychomotor learning.

We know from learning in real environments that the likeness of the instructor to the learner can moderate learning effects [46]. The neuroscientific basis for observational learning is thought to lie with mirror neurons, which activate both when performing a skill as well as when observing it [31]. If an instructor looks similar or is familiar to a learner, this has the potential to increase mirror neuron activity and therefore enhance learning [12]. This has been utilised in *feedforward learning*, where a learner observes a model of themselves, typically through video which has been edited to give the impression of a better skill performance [14]. This has been shown to result in rapid improvements in performance across a range of activities, including motor tasks [11]. In consequence, making instructor avatars look like the learner is promising and has been studied for various forms of learning and behaviour change in virtual environments [18]. However, apart from a comparison of skilled versus novice self-avatars [18], and a small exploratory study by Fitton et al. [17], it has not yet been investigated for learning psychomotor tasks. How much similarity is necessary to achieve meaningful enhancements in learning has not been investigated yet. This raises the question of whether minimal avatar customisation can enhance learning of a psychomotor skill, and if so, do the effects of minimal avatar customisation differ when learning from a screen compared to learning in VR?

We demonstrate how minimal customisation of an instructor avatar can enhance learning of a complex psychomotor task – modern dancing. Rather than trying to make the instructor a digital clone of the learner, we customise only three basic properties to make it more similar to the learner: gender, skin colour and hair colour. Such minimal customisation mitigates concerns of highly-similar avatars: it is very easy to implement, requires minimal effort from the user, avoids negative effects of poorly designed avatars and strong self-similarity, and helps preserve the learner’s privacy. Basic avatar customisation of gender, skin colour, and hair colour can be done in a matter of seconds and is commonplace in many virtual environments [42]. We demonstrate the effects of minimal avatar customisation in both a screen-based and immersive VR virtual environment.

3 PRESENCE IN VR - REALISM AND FIELD OF VIEW AFFECT PRESENCE IN VR BUT NOT THE WAY YOU THINK [27]

Head mounted display (HMD) technology can immerse a user’s senses in the virtual world and help induce feelings of presence. According to Witmer and Singer [52], presence as a construct refers to the user’s subjective feeling of actually being in a place or environment, even when they are situated in another. However, in the context of experiencing virtual environments (VEs), presence refers to experiencing the VE rather than the actual physical one [52]. In contrast, immersion refers to an objective property of a system and the extent to which it can engage the user’s sensorimotor channels and perception of the VE [47]. Immersion is therefore affected by the technical characteristics of the VR setup including characteristics of both the software, such as realism of the VE, and the hardware, such as the field of view of the HMD [6]. Critically, the level of immersion provided by the system may affect the illusion of being in the virtual world, and the process by which presence is created has been described as a ‘negotiation’ between the user and the technical, or immersive, qualities of a VR system [26].

Despite technological advances there are certain characteristics, such as VE realism and HMD FoV, that compete for limited computational resources, and designers of VR experiences still need to prioritise certain technical improvements over others when designing for the masses. This raises questions about how technical factors interact with each other to affect presence, and to date there has been little to no systematic combinatorial investigation. In addition to the immersive quality of the VR setup, research has shown how human factors can affect a user’s presence. The emotions felt by the user [20] and their perceived agency within the VE [21] have been identified as important human factors that affect presence, and there are complex interactions that occur during this process [29]. Emotions that rate highly on arousal, and in particular fear, appear to hold a special place in the formation of presence [20], having a strong evolutionary importance [38]. Furthermore, eliciting a sense of fear is at the centre of a wide range of VR applications, including therapeutic interventions [40], training for crisis management [19], and desensitisation to phobia-inducing stimuli [28], and is even commonly used as a game mechanic in popular VR games [34].

We systematically investigated the effects on presence of important technical factors in the wider context of human factors. That is, we provide insights into a) interactions between the two technical factors realism and field of view, b) interactions between these technical factors and the two human factors agency and fear, and c) the importance of the technical and human factors in the formation of presence relative to each other. We conducted a large-scale study with 360 participants exploring the formation of presence in VR, by systematically varying the technical factors visual realism (high/low) and FoV (high/low), as well as the human factors emotion (focusing on fear) and agency (yes/no), yielding $2 \times 2 \times 2 \times 2 = 16$ between-group conditions. In summary, we make the following contributions: (i) evidence that visual realism and FoV do not affect presence directly; (ii) evidence that the effects of visual realism and FoV on presence are moderated by induced fear and perceived agency, respectively; and (iii) the TAP-Fear model, describing how technical and human factors work together in presence formation.

4 EXERGAMING IN VR - RACE YOURSELVES: A LONGITUDINAL EXPLORATION OF SELF-COMPETITION BETWEEN PAST, PRESENT, AND FUTURE PERFORMANCES IN A VR EXERGAME [37]

Sedentary behaviour has detrimental consequences for health and well-being. Not only is a lack of physical activity strongly associated with obesity [50] and sub-optimal mental health [43], but it is also one of the leading causes of preventable death worldwide [35]. It is estimated that about 30% of adults and 80% of adolescents fail to meet public health guidelines for physical activity [22] all while dropout rates from exercise programs continue to hover around 50% [10]. One of the strongest predictors of exercise adherence is intrinsic motivation [2], i.e. doing an activity for its inherent satisfaction and enjoyment. Alongside perceived lack of time, not enjoying exercise is commonly cited as the most important barrier to long-term exercise adherence [45]. Making exercise intrinsically motivating and time-efficient is, therefore, an important step towards tackling the global epidemic of sedentary behaviour.

Participating in racing sports can be a thrilling experience for many athletes, and the euphoric experience of winning a tough race is often accompanied by a boost in self-esteem, pride, and confidence [51]. However, as exciting as racing can be to athletes, its competitive nature is not appealing to everyone. There exists extensive literature showing that competing against others can have detrimental effects on intrinsic motivation, especially for less fit, less competitive individuals [7]. Cognitive evaluation theory [44] suggests that low perceived competence in an activity translates into diminished intrinsic motivation [8], i.e. it is difficult for someone to enjoy an activity they do not feel they are good at. Given these tough psychological obstacles, can we make the positive racing experience that athletes find so enticing accessible to less fit and less competitive individuals?

Exergaming, which makes exercise more intrinsically motivating through gamification [39], yields a potential solution: self-competition. Self-competition is a common feature of many racing video games [5] where players race against an opponent that is a replay or “ghost” of one of their past performances. The problem with existing forms of self-competitive exergaming is that players only compete against a single past performance [15], which limits the richness of the racing experience. Unlike one-on-one competition, being surrounded by and competing against a spectrum of differently skilled competitors creates a shared experience that is a key ingredient of many exciting races both in video games and in real life [23]. To bridge the gap between self-competition and such a rich racing experience, we propose a novel approach in which players race against their own *performance history* as well as a model of their future performance using multiple ghost avatars (Figure 1c), creating a new form of competition that has not been previously explored.

We decided to implement our “multi-ghost” approach in a high-intensity cycling exergame using VR to evoke the same visceral sensations felt during a real racing event [23]. Having players race against their past and future selves in an immersive virtual environment offers the following potential advantages:

Fairness: Because competitors are all generated from the player’s own performances, competition becomes inherently fair and personalised to each player’s fitness ability. This helps to strike the right balance between skill and challenge, which facilitates flow [41] and intrinsic motivation [30].

Feedforward is a psychological training technique used to achieve rapid performance improvement by establishing an image of an ideal future self that the present self can learn from [13]. Allowing players to compete against a “future ghost”, which is an enhanced self-model of their best performance to date, can motivate them to exercise at a higher intensity without negatively impacting their intrinsic motivation [4]. This would help them achieve their fitness goals more quickly.

Self-Efficacy, the confidence in one’s ability to carry out a task under challenging circumstances [3], is an important predictor of exercise adherence [9]. Self-competitive exergaming can avoid the common detrimental effects of conventional competition on self-efficacy [4] and beating your own ghosts could cultivate perceived competence, an important contributor to self-efficacy [1].

Adaptability: The latest ghosts reflect the player’s latest performances, so as players improve, so do their competitors. This adaptive approach to increasing competitive pressure encourages players to continually improve their performance, but in a way that always maintains a balance between the challenge and the player’s current skill [25].

Replayability: After each game session, a new ghost is generated and added to the set of competitors. Competing against an extra ghost every time the game is played again ensures that the competitive dynamics are never the same. This helps increase the replay value of the exergame [32] and makes it more likely to sustain interest over the longer term, hence addressing a common limitation of existing exergames [36].

REFERENCES

- [1] Terrence B Abney. 2007. The Importance of Perceived Competence in Fitness/Exercise Programming. *The Sport Journal* 5 (2007).
- [2] Konstantinos Alexandris, Charalambos Tsozatzoudis, and George Grouios. 2002. Perceived Constraints on Recreational Sport Participation: Investigating Their Relationship With Intrinsic Motivation, Extrinsic Motivation and Amotivation. *Journal of Leisure Research* 34, 3 (2002), 233.
- [3] Albert Bandura. 2010. Self-Efficacy. *The Corsini Encyclopedia of Psychology* (2010), 1–3.
- [4] Soumya C Barathi, Daniel J Finnegan, Matthew Farrow, Alexander Whaley, Pippa Heath, Jude Buckley, Peter W Dowrick, Burkhard C Wuensche, James LJ Bilzon, Eamonn O'Neill, et al. 2018. Interactive Feedforward for Improving Performance and Maintaining Intrinsic Motivation in VR Exergaming. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. ACM, 408–421.
- [5] Giant Bomb contributors. 2019. Ghost Racer (Concept). Wiki. Retrieved September 15, 2019 from <https://www.giantbomb.com/ghost-racer/3015-3097/>.
- [6] James J Cummings and Jeremy N Bailenson. 2016. How immersive is enough? A meta-analysis of the effect of immersive technology on user presence. *Media Psychology* 19, 2 (2016), 272–309.
- [7] Edward L Deci and Bradley C Olson. 2012. Motivation and Competition: Their Role in Sports. In *Sports, Games, and Play*. Psychology Press, 93–120.
- [8] Edward L Deci and Richard M Ryan. 2010. Intrinsic Motivation. *The Corsini Encyclopedia of Psychology* (2010), 1–2.
- [9] Raymond Desharnais, Jacques Bouillon, and Gaston Godin. 1986. Self-Efficacy and Outcome Expectations as Determinants of Exercise Adherence. *Psychological Reports* 59, 3 (1986), 1155–1159. <https://doi.org/10.2466/pr0.1986.59.3.1155> arXiv:<https://doi.org/10.2466/pr0.1986.59.3.1155>
- [10] Rod K Dishman. 1991. Increasing and Maintaining Exercise and Physical Activity. *Behavior Therapy* 22, 3 (1991), 345–378.
- [11] Peter W. Dowrick. 1999. A review of self modeling and related interventions. *Applied and Preventive Psychology* 8, 1 (Dec. 1999), 23–39. [https://doi.org/10.1016/S0962-1849\(99\)80009-2](https://doi.org/10.1016/S0962-1849(99)80009-2)

- [12] Peter W. Dowrick. 2012. Self model theory: learning from the future. *WIREs Cognitive Science* 3, 2 (2012), 215–230. <https://doi.org/10.1002/wcs.1156> eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/wcs.1156>.
- [13] Peter W Dowrick. 2012. Self Model Theory: Learning From the Future. *Wiley Interdisciplinary Reviews: Cognitive Science* 3, 2 (2012), 215–230.
- [14] Peter W. Dowrick, Weol Soon Kim-Rupnow, and Thomas J. Power. 2006. Video Feedforward for Reading. *J Spec Educ* 39, 4 (Feb. 2006), 194–207. <https://doi.org/10.1177/00224669060390040101> Publisher: SAGE Publications Inc.
- [15] Matthew Farrow, Christof Lutteroth, Peter C Rouse, and James LJ Bilzon. 2018. Virtual-Reality Exergaming Improves Performance During High-Intensity Interval Training. *European Journal of Sport Science* (2018), 1–9.
- [16] Isabel Fitton, Christopher Clarke, Jeremy Dalton, Michael Proulx, and Christof Lutteroth. 2023. Dancing with the Avatars: Minimal Avatar Customisation Enhances Learning in a Psychomotor Task. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*.
- [17] Isabel Sophie Fitton, Jeremy Dalton, Michael J Proulx, and Christof Lutteroth. 2022. Dancing with the Avatars: Feedforward Learning from Self-Avatars. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts* (New Orleans, LA, USA) (CHI EA '22). Association for Computing Machinery, New York, NY, USA, Article 458, 8 pages. <https://doi.org/10.1145/3491101.3519732>
- [18] Cornelia Frank, Felix Hülsman, Thomas Waltemate, David J Wright, Daniel L Eaves, Adam Bruton, Mario Botsch, and Thomas Schack. 2022. Motor imagery during action observation in virtual reality: the impact of watching myself performing at a level I have not yet achieved. *International Journal of Sport and Exercise Psychology* (2022), 1–27.
- [19] Luciano Gamberini, Luca Chittaro, Anna Spagnoli, and Claudio Carlesso. 2015. Psychological response to an emergency in virtual reality: Effects of victim ethnicity and emergency type on helping behavior and navigation. *Computers in Human Behavior* 48 (2015), 104–113.
- [20] Daniel Gromer, Max Reinke, Isabel Christner, and Paul Pauli. 2019. Causal interactive links between presence and fear in virtual reality height exposure. *Frontiers in psychology* 10 (2019), 141.
- [21] Olga Gutiérrez-Martínez, José Gutiérrez-Maldonado, and Desirée Loreto-Quijada. 2011. Control over the virtual environment influences the presence and efficacy of a virtual reality intervention on pain.
- [22] Pedro C Hallal, Lars Bo Andersen, Fiona C Bull, Regina Guthold, William Haskell, Ulf Ekelund, Lancet Physical Activity Series Working Group, et al. 2012. Global Physical Activity Levels: Surveillance Progress, Pitfalls, and Prospects. *The Lancet* 380, 9838 (2012), 247–257.
- [23] Joshua C Haller, Young H Jang, Jack Haller, Lindsay Shaw, and Burkhard C Wünsche. 2019. HIIT The Road: Using Virtual Spectator Feedback in HIIT-based Exergaming. In *Proceedings of the Australasian Computer Science Week Multiconference*. ACM, 47.
- [24] CM Heyes and CL Foster. 2002. Motor learning by observation: evidence from a serial reaction time task. *The Quarterly Journal of Experimental Psychology Section A* 55, 2 (2002), 593–607.
- [25] Susan A Jackson and Mihaly Csikszentmihalyi. 1999. *Flow in Sports*. Human Kinetics.
- [26] Charlene Jennett, Anna L Cox, Paul Cairns, Samira Dhoparee, Andrew Epps, Tim Tijs, and Alison Walton. 2008. Measuring and defining the experience of immersion in games. *International Journal of Human-Computer Studies* 66, 9 (2008), 641–661.
- [27] Crescent Jicol, Christopher Clarke, Emilia Tor, Rebecca Dakin, Tom Lancaster, Sze Tung Chang, Karin Petrini, Eamonn O'Neill, Michael Proulx, and Christof Lutteroth. 2023. Realism and Field of View Affect Presence in VR but Not the Way You Think. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*.
- [28] Crescent Jicol, Julia Feltham, Jinha Yoon, Michael J Proulx, Eamonn O'Neill, and Christof Lutteroth. 2022. Designing and Assessing a Virtual Reality Simulation to Build Resilience to Street Harassment. In *CHI Conference on Human Factors in Computing Systems*. 1–14.
- [29] Crescent Jicol, Chun Hin Wan, Benjamin Doling, Caitlin H Illingworth, Jinha Yoon, Charlotte Headey, Christof Lutteroth, Michael J Proulx, Karin Petrini, and Eamonn O'Neill. 2021. Effects of emotion and agency on presence in virtual reality. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [30] Johannes Keller and Herbert Bless. 2008. Flow and Regulatory Compatibility: An Experimental Approach to the Flow Model of Intrinsic Motivation. *Personality and Social Psychology Bulletin* 34, 2 (2008), 196–209.
- [31] James M Kilner and Roger N Lemon. 2013. What we know currently about mirror neurons. *Current biology* 23, 23 (2013), R1057–R1062.
- [32] Wolfgang Kramer. 2000. What Makes a Game Good. *Game & Puzzle Design, vol. 1, no. 2, 2015 (Colour)* (2000), 84.
- [33] Serena Lee-Cultura, Kshitij Sharma, Sofia Papavaslopoulou, Symeon Retalis, and Michail Giannakos. 2020. Using sensing technologies to explain children's self-representation in motion-based educational games. In *Proceedings of the interaction design and children conference*. 541–555.
- [34] Jih-Hsuan Tammy Lin. 2017. Fear in virtual reality (VR): Fear elements, coping reactions, immediate and next-day fright responses toward a survival horror zombie virtual reality game. *Computers in Human Behavior* 72 (2017), 350–361.
- [35] Alan D Lopez, Colin D Mathers, Majid Ezzati, Dean T Jamison, and Christopher JL Murray. 2006. Global and Regional Burden of Disease and Risk Factors, 2001: Systematic Analysis of Population Health Data. *The Lancet* 367, 9524 (2006), 1747–1757.
- [36] Andrew Macvean and Judy Robertson. 2013. Understanding Exergame Users' Physical Activity, Motivation and Behavior Over Time. In *Proc. CHI ACM*, 1251–1260.
- [37] Alexander Michael and Christof Lutteroth. 2020. Race yourselves: A longitudinal exploration of self-competition between past, present, and future performances in a vr exergame. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–17.
- [38] Maarten Milders, Arash Sahraie, Sarah Logan, and Niamh Donnellon. 2006. Awareness of faces is modulated by their emotional meaning. *Emotion* 6, 1 (2006), 10.
- [39] Yoonsin Oh and Stephen Yang. 2010. Defining exergames & exergaming. *Proceedings of meaningful play* 2010 (2010), 21–23.
- [40] Mi Jin Park, Dong Jun Kim, Unjoo Lee, Eun Jin Na, and Hong Jin Jeon. 2019. A literature overview of virtual reality (VR) in treatment of psychiatric disorders: recent advances and limitations. *Frontiers in psychiatry* 10 (2019), 505.
- [41] R Lingyak Petosa and Brian Holtz. 2013. Flow for Exercise Adherence: Testing an Intrinsic Model of Health Behavior. *American Journal of Health Education* 44, 5 (2013), 273–277.
- [42] Katherine M. Rahill and Marc M. Sebrechts. 2021. Effects of Avatar player-similarity and player-construction on gaming performance. *Computers in Human Behavior Reports* 4 (2021), 100131. <https://doi.org/10.1016/j.chbr.2021.100131>
- [43] James E Rohrer, J Rush Pierce Jr, and Claudia Blackburn. 2005. Lifestyle and Mental Health. *Preventive Medicine* 40, 4 (2005), 438–443.
- [44] Richard M Ryan. 1982. Control and Information in the Intrapersonal Sphere: An Extension of Cognitive Evaluation Theory. *Journal of Personality and Social Psychology* 43, 3 (1982), 450–461.
- [45] James F Sallis, Melbourne F Hovell, and C Richard Hofstetter. 1992. Predictors of Adoption and Maintenance of Vigorous Physical Activity in Men and Women. *Preventive Medicine* 21, 2 (1992), 237–251.
- [46] Dale H Schunk. 1987. Peer models and children's behavioral change. *Review of educational research* 57, 2 (1987), 149–174.
- [47] Mel Slater. 2018. Immersion and the illusion of presence in virtual reality. *British Journal of Psychology* 109, 3 (2018), 431–433.
- [48] Katherine E Sukel, Richard Catrambone, Irfan Essa, and Gabriel Brostow. 2003. Presenting movement in a computer-based dance tutor. *International Journal of Human-Computer Interaction* 15, 3 (2003), 433–452.
- [49] Milka Trajkova, A'eshah Alhakamy, Francesco Cafaro, Rashmi Mallappa, and Sreekanth R Kankara. 2020. Move your body: Engaging museum visitors with human-data interaction. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [50] Mark Stephen Tremblay, Rachel Christine Colley, Travis John Saunders, Genevieve Nissa Healy, and Neville Owen. 2010. Physiological and Health Implications of a Sedentary Lifestyle. *Applied Physiology, Nutrition, and Metabolism* 35, 6 (2010), 725–740.
- [51] George V Wilson and John H Kerr. 1999. Affective Responses to Success and Failure: A Study of Winning and Losing in Competitive Rugby. *Personality and Individual Differences* 27, 1 (1999), 85–99.
- [52] Bob G Witmer and Michael J Singer. 1998. Measuring presence in virtual environments: A presence questionnaire. *Presence* 7, 3 (1998), 225–240.